

DATA STRUCTURE

LAB PROGRAMS

11. a) Stack – Push operation using linked list

```
#include <stdio.h>

#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *top, *newNode, *temp;
    int value;
    top = (struct Node *)malloc(sizeof(struct Node));
    top->data = 30;
    top->next = (struct Node *)malloc(sizeof(struct Node));
    top->next->data = 20;
    top->next->next = (struct Node *)malloc(sizeof(struct Node));
    top->next->next->data = 10;
    top->next->next->next = NULL;
    printf("Enter value to push: ");
    scanf("%d", &value);
    newNode = (struct Node *)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = top;
```

```

top = newNode;
printf("Stack after Push:\n");
temp = top;
while(temp != NULL)
{
    printf("%d\n", temp->data);
    temp = temp->next;
}
return 0;
}

```

Output

```

Enter value to push: 38
Stack after Push:
38
30
20
10

=== Code Execution Successful ===

```

b) Stack – Pop operation using linked list

```

#include <stdio.h>

#include <stdlib.h>

struct Node

```

```

{
    int data;

    struct Node *next;
};

int main()
{
    struct Node *top, *temp;

    top = (struct Node *)malloc(sizeof(struct Node));
    top->data = 30;
    top->next = (struct Node *)malloc(sizeof(struct Node));
    top->next->data = 20;
    top->next->next = (struct Node *)malloc(sizeof(struct Node));
    top->next->next->data = 10;
    top->next->next->next = NULL;

    if(top == NULL)
    {
        printf("Stack Underflow");
    }
    else
    {
        temp = top;

        printf("Deleted Element = %d\n", top->data);

        top = top->next;

        free(temp);
    }

    printf("Stack after Pop:\n");

    temp = top;

    while(temp != NULL)
    {

```

```
    printf("%d\n", temp->data);  
    temp = temp->next;  
}  
return 0;  
}
```

Output

Deleted Element = 30

Stack after Pop:

20

10

=== Code Execution Successful ===

c) Stack – Display operation using linked list

```
#include <stdio.h>  
#include <stdlib.h>  
  
struct Node  
{  
    int data;  
    struct Node *next;  
};  
  
int main()  
{
```

```
struct Node *top, *temp;

top = (struct Node *)malloc(sizeof(struct Node));

top->data = 30;

top->next = (struct Node *)malloc(sizeof(struct Node));

top->next->data = 20;

top->next->next = (struct Node *)malloc(sizeof(struct Node));

top->next->next->data = 10;

top->next->next->next = NULL;

if(top == NULL)

{

    printf("Stack is Empty");

}

else

{

    printf("Stack Elements:\n");

    temp = top;

    while(temp != NULL)

    {

        printf("%d\n", temp->data);

        temp = temp->next;

    }

}

return 0;

}
```

Output

Stack Elements:

30

20

10

=== Code Execution Successful ===